Multiplying and Dividing Fractions

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS numerator *(taurunga)* · denominator *(tauraro)* · reciprocal · cancel · mixed number *(tau whakauru)*

A Multiply, cancelling first ●○○

Remember: Multiply the tops and multiply the bottoms. Cancel a common factor from a top and a bottom before you multiply to keep the numbers small.

1. Work out. Give each answer in simplest form.

a) $\frac{3}{4} \times \frac{8}{9} =$ _____ **b)** $\frac{2}{5} \times \frac{5}{6} =$ _____ **c)** $\frac{5}{6} \times \frac{3}{10} =$ _____ **d)** $\frac{4}{7} \times \frac{7}{8} =$ _____

e) $\frac{9}{10} \times \frac{5}{12} =$ _____ **f)** $\frac{7}{8} \times \frac{4}{21} =$ _____ **g)** $\frac{3}{5} \times \frac{10}{9} =$ _____ **h)** $\frac{11}{12} \times \frac{6}{11} =$ _____

B Mixed numbers ●●○

Remember: Change each mixed number to an improper fraction first. To divide, multiply by the reciprocal of the second fraction.

2. Work out. Give each answer as a mixed number or a fraction in its simplest form.

a) $1\frac{1}{2} \times 2\frac{2}{3}$ **b)** $2\frac{1}{4} \times 1\frac{3}{5}$ **c)** $3\frac{1}{3} \times 1\frac{1}{2}$

d) $1\frac{3}{4} \times 2\frac{2}{7}$ **e)** $2\frac{1}{2} \div 1\frac{1}{4}$ **f)** $3\frac{3}{4} \div 1\frac{1}{2}$

g) $1\frac{2}{3} \div 2\frac{1}{2}$ **h)** $4\frac{1}{2} \div 3\frac{3}{8}$ **i)** $2\frac{2}{5} \div 1\frac{1}{5}$

C Negative fractions ●●○

Remember: Decide the sign first: two signs the same give a positive answer, two different signs give a negative answer. Then multiply or divide the fractions as usual.

3. Work out. Give each answer in simplest form.

a) $-\frac{2}{3} \times \frac{3}{4}$ **b)** $\frac{1}{2} \times \left(-\frac{4}{5}\right)$ **c)** $-\frac{3}{5} \times \left(-\frac{5}{6}\right)$

d) $-\frac{7}{8} \times 4$ **e)** $-\frac{1}{2} \div \frac{3}{4}$ **f)** $\frac{2}{3} \div \left(-\frac{4}{9}\right)$

g) $-\frac{3}{4} \div \left(-\frac{1}{2}\right)$ **h)** $-\frac{5}{6} \div 10$ **i)** $-\frac{9}{10} \times \left(-\frac{5}{3}\right)$

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D Negative mixed numbers ●●●

⚠ **Watch out:** A negative mixed number is all negative: $-1\frac{1}{2}$ means $-\frac{3}{2}$. Change it to an improper fraction with the sign in front.

4. Work out. Give each answer as a mixed number or a fraction in its simplest form.

a) $-1\frac{1}{2} \times 2\frac{2}{3}$ **b)** $2\frac{1}{4} \times \left(-1\frac{1}{3}\right)$ **c)** $-1\frac{3}{5} \times \left(-2\frac{1}{2}\right)$

d) $-2\frac{1}{4} \div 1\frac{1}{2}$ **e)** $1\frac{1}{3} \div \left(-2\frac{2}{3}\right)$ **f)** $-3\frac{1}{2} \div \left(-1\frac{3}{4}\right)$

E Problems ●●●

5. A rectangle is $2\frac{1}{2}$ m long and $1\frac{3}{4}$ m wide. Find its area.

6. A barrel holds $6\frac{3}{8}$ litres of juice. It is poured into bottles that each hold $\frac{3}{4}$ of a litre. How many bottles can be filled completely?

7. A cyclist rides $2\frac{1}{4}$ km in $\frac{3}{4}$ of an hour. What is the speed in km/h? (Speed = distance ÷ time.)

8. A recipe needs $2\frac{2}{3}$ cups of flour. Hana makes $1\frac{1}{2}$ times the recipe. How much flour does she need?

★ Extension

9. Work out $\left(1\frac{1}{2}\right)^2$.

10. Work out $\sqrt{2\frac{1}{4}}$.

11. Work out $\left(-\frac{2}{3}\right)^3$.

12. Work out $\frac{1}{2} \div \frac{1}{3} \div \frac{1}{4}$. Work from left to right.

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Multiplying and Dividing Fractions

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS numerator *(taurunga)* · denominator *(tauraro)* · reciprocal · cancel · mixed number *(tau whakauru)*

A Multiply, cancelling first ●○○

Remember: Multiply the tops and multiply the bottoms. Cancel a common factor from a top and a bottom before you multiply to keep the numbers small.

1. Work out. Give each answer in simplest form.

a) $\frac{2}{3} \times \frac{9}{10} =$ _____ **b)** $\frac{3}{8} \times \frac{4}{9} =$ _____ **c)** $\frac{5}{7} \times \frac{7}{10} =$ _____ **d)** $\frac{4}{5} \times \frac{5}{12} =$ _____

e) $\frac{7}{9} \times \frac{3}{14} =$ _____ **f)** $\frac{5}{6} \times \frac{9}{20} =$ _____ **g)** $\frac{2}{7} \times \frac{7}{8} =$ _____ **h)** $\frac{11}{15} \times \frac{5}{22} =$ _____

B Mixed numbers ●●○

Remember: Change each mixed number to an improper fraction first. To divide, multiply by the reciprocal of the second fraction.

2. Work out. Give each answer as a mixed number or a fraction in its simplest form.

a) $1\frac{1}{3} \times 2\frac{1}{4}$ **b)** $2\frac{1}{2} \times 1\frac{3}{5}$ **c)** $3\frac{1}{2} \times 1\frac{1}{7}$

d) $1\frac{2}{3} \times 2\frac{1}{4}$ **e)** $2\frac{1}{4} \div 1\frac{1}{2}$ **f)** $3\frac{1}{3} \div 1\frac{2}{3}$

g) $1\frac{3}{4} \div 2\frac{1}{3}$ **h)** $4\frac{1}{5} \div 2\frac{4}{5}$ **i)** $2\frac{5}{8} \div 1\frac{1}{4}$

C Negative fractions ●●○

Remember: Decide the sign first: two signs the same give a positive answer, two different signs give a negative answer. Then multiply or divide the fractions as usual.

3. Work out. Give each answer in simplest form.

a) $-\frac{3}{4} \times \frac{2}{5}$ **b)** $\frac{1}{3} \times \left(-\frac{6}{7}\right)$ **c)** $-\frac{2}{5} \times \left(-\frac{5}{8}\right)$

d) $-\frac{5}{6} \times 3$ **e)** $-\frac{1}{3} \div \frac{2}{5}$ **f)** $\frac{3}{4} \div \left(-\frac{9}{8}\right)$

g) $-\frac{2}{5} \div \left(-\frac{3}{10}\right)$ **h)** $-\frac{7}{8} \div 14$ **i)** $-\frac{8}{9} \times \left(-\frac{3}{4}\right)$

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D Negative mixed numbers ●●●

⚠ **Watch out:** A negative mixed number is all negative: $-1\frac{1}{2}$ means $-\frac{3}{2}$. Change it to an improper fraction with the sign in front.

4. Work out. Give each answer as a mixed number or a fraction in its simplest form.

a) $-1\frac{1}{3} \times 2\frac{1}{4}$ **b)** $2\frac{1}{2} \times \left(-1\frac{1}{5}\right)$ **c)** $-1\frac{2}{5} \times \left(-2\frac{1}{2}\right)$

d) $-2\frac{1}{2} \div 1\frac{1}{4}$ **e)** $1\frac{1}{4} \div \left(-2\frac{1}{2}\right)$ **f)** $-3\frac{3}{4} \div \left(-1\frac{1}{2}\right)$

E Problems ●●●

5. A rectangle is $3\frac{1}{2}$ m long and $1\frac{1}{4}$ m wide. Find its area.

6. A barrel holds $5\frac{1}{4}$ litres of juice. It is poured into bottles that each hold $\frac{3}{8}$ of a litre. How many bottles can be filled completely?

7. A cyclist rides $3\frac{3}{4}$ km in $\frac{3}{4}$ of an hour. What is the speed in km/h? (Speed = distance ÷ time.)

8. A recipe needs $1\frac{3}{5}$ cups of flour. Hana makes $2\frac{1}{2}$ times the recipe. How much flour does she need?

★ Extension

9. Work out $\left(2\frac{1}{2}\right)^2$.

10. Work out $\sqrt{6\frac{1}{4}}$.

11. Work out $\left(-\frac{3}{4}\right)^3$.

12. Work out $\frac{2}{3} \div \frac{1}{2} \div \frac{1}{6}$. Work from left to right.

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